Automated Experiment Report

Run folder: run_20260418_214955

Prepared on 2026-04-18 20:44:39. This report was created automatically from saved run artifacts using deterministic Python rules only.

Model family Direct Hurdle Tabular	**Horizons** 24, 48
Platform Linux-6.6.113+-x86_64-with-glibc2.35	**XGBoost device** cuda
Classifier configs 2	**Regressor configs** 2
Pairs per horizon 4	**Total fold fits** 8

Executive summary

- The run archived 142 tracked artifacts.
- Best 24h pair: h24__clf_xgbc01__reg_xgbr01.
- Best 48h pair: h48__clf_xgbc01__reg_xgbr01.
- For horizon 24h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 82.8958; Improved by 3.8139 (4.40%). Stage-1 ROC AUC was 0.8965, indicating strong discrimination.
- For horizon 48h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 64.6970; Improved by 2.0198 (3.03%). Stage-1 ROC AUC was 0.8744, indicating strong discrimination.
- Submission-style CSVs were exported from: Best validation candidate.
- Demo-only evaluation files are present. They are useful for sanity checks only and are not the hidden challenge evaluation.

Configuration and modeling decisions

- Architecture: the run uses a two-stage direct hurdle pipeline ('direct_hurdle_tabular'). Stage 1 predicts whether outage is non-zero; Stage 2 predicts outage size conditional on positive outcomes only.
- Combination rule: the final expected outage equals stage-1 non-zero probability times the stage-2 positive prediction.
- Target transform: Stage 2 uses log1p on positive outages before fitting the regressor.
- Forecast horizons: 24h, 48h.
- Forecast-origin sampling: every eligible hour was used.
- Minimum training history before the first fold: 21 days (504 h).
- Maximum training history: capped at 28 days (672 h).

- Temporal validation design: manual aligned splits were specified - 24h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-12 00:00:00; 48h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-12 00:00:00.
- Target lag features: 24, 48 hours.
- Target rolling windows: 6 hours.
- Weather lag features: 24, 48 hours.
- Weather rolling windows: 6 hours.
- Weather screening: up to 10 weather variables were allowed; 10 were retained in this run.
- Weather outlier handling: enabled with 'winsorize_iqr', so extreme weather values were clipped rather than turned into missing values.
- Execution hardware: GPU acceleration was available; XGBoost used device 'cuda'.
- Prediction export policy: submission CSVs were generated from Best validation candidate.

Search space summary

component	families	config_count
Classifier	logistic_enet, xgb_classifier	2
Regressor	elasticnet_regression, xgb_regressor	2

Run inventory

The table below shows the top-level artifact folders detected in the run. The checkpoints folder is runtime infrastructure and may be intentionally absent from shared GitHub uploads.

subfolder	present	file_count	note
tables	Yes	33	
predictions	Yes	12	
models	Yes	40	
metrics	Yes	28	
plots	Yes	17	
checkpoints	Yes	10	Runtime-only resume artifacts; may be omitted from shared GitHub uploads.

Temporal evaluation design

The backtest table below maps the training and validation windows used for each horizon and fold.

horizon	fold_id	val_origin_idx	val_origin_timestamp	train_origin_start_idx	train_origin_start_timestamp	train_origin_max_idx	train_origin_max_timestamp	validation_window_start	validation_window_end
24	1	1,728	2023-06-12	1,464	2023-06-01	1,704	2023-06-11	2023-06-12 01:00:00	2023-06-13
48	1	1,728	2023-06-12	1,464	2023-06-01	1,680	2023-06-10	2023-06-12 01:00:00	2023-06-14

Feature frame and data overview

metric	value
rows_in_model_df	179,363
base_numeric_features	74

metric	value
categorical_features	1.0000
feature_warmup_hours	48
max_horizon	48
mean_outage	45.2483
median_outage	0.0000
std_outage	452.2684
max_outage	23,346
zero_rate	0.7047
nonzero_rate	0.2953

Selected weather features retained in this run: 10.

feature	abs_corr_with_outage	signed_corr_with_outage
gh_4	0.1299	0.1299
cape_1	0.0982	0.0982
pwat	0.0885	0.0885
sh2	0.0832	0.0832
mslma	0.0758	-0.0758
sdlwrf	0.0707	0.0707
mcc	0.0662	0.0662
d2m	0.0650	0.0650
cape	0.0641	0.0641
lcc	0.0554	0.0554

Candidate ranking by horizon

Rows are ranked within each horizon using average county RMSE expected first and overall RMSE expected second.

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
1	24	xgb_classifier	xgb_regressor	82.8958	255.2574	3.8139	18.0962	0.8965	425.4050	h24__clf_xgbc01__reg_xgbr01
2	24	logistic_enet	xgb_regressor	83.6580	254.9840	3.0517	18.3696	0.8976	425.4050	h24__clf_logen01__reg_xgbr01
3	24	xgb_classifier	elasticnet_regression	84.3897	262.1841	2.3200	11.1695	0.8965	435.9580	h24__clf_xgbc01__reg_enet01
4	24	logistic_enet	elasticnet_regression	84.9082	261.9930	1.8015	11.3606	0.8976	435.9580	h24__clf_logen01__reg_enet01
1	48	xgb_classifier	xgb_regressor	64.6970	186.3607	2.0198	10.3699	0.8744	321.7745	h48__clf_xgbc01__reg_xgbr01

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
2	48	logistic_enet	xgb_regressor	65.2716	185.8631	1.4451	10.8676	0.8754	321.7745	h48__clf_logen01__reg_xgbr01
3	48	xgb_classifier	elasticnet_regression	65.5797	189.2101	1.1370	7.5206	0.8744	328.1793	h48__clf_xgbc01__reg_enet01
4	48	logistic_enet	elasticnet_regression	66.4344	189.1047	0.2824	7.6260	0.8754	328.1793	h48__clf_logen01__reg_enet01

Winning pairs by horizon

horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage1_f1	mean_stage2_positive_rmse	pair_tag
24	xgb_classifier	xgb_regressor	82.8958	255.2574	3.8139	18.0962	0.8965	0.7090	425.4050	h24__clf_xgbc01__reg_xgbr01
48	xgb_classifier	xgb_regressor	64.6970	186.3607	2.0198	10.3699	0.8744	0.6470	321.7745	h48__clf_xgbc01__reg_xgbr01

Horizon 24h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 82.8958 versus a zero baseline of 86.7097. Improved by 3.8139 (4.40%).

Horizon 48h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 64.6970 versus a zero baseline of 66.7168. Improved by 2.0198 (3.03%).

Stage-specific summaries

Stage 1 classifiers

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
24	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8965	0.8659	0.1237	0.7090
24	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.8976	0.8543	0.1236	0.7590

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
48	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8744	0.8266	0.1300	0.6470
48	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.8754	0.8237	0.1372	0.7285

Stage 2 regressors

horizon	regressor_name	regressor_params	mean_stage2_positive_rmse	mean_stage2_positive_mae	mean_stage2_positive_rows_train
24	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	425.4050	162.3080	140,860
24	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	435.9580	167.8878	140,860
48	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	321.7745	101.9745	252,291
48	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	328.1793	106.1644	252,291

Submission and refit artifacts

Submission model summary

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
24	Best validation candidate	h24__clf_xgbc01__reg_xgbr01	1	xgb_classifier	xgb_regressor	82.8958	255.2574

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
48	Best validation candidate	h48__clf_xgbc01__reg_xgbr01	1	xgb_classifier	xgb_regressor	64.6970	186.3607

Final refit summary

horizon	bundle_source	pair_tag	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected
24	Final refit	h24__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	82.8958	255.2574
48	Final refit	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	64.6970	186.3607

Prediction exports

file	rows	prediction_column	min	max	mean	std
two_stage_pred_24h.csv	1,992	pred	0.1296	174.5334	6.9448	23.2673
two_stage_pred_48h.csv	3,984	pred	0.2463	203.4567	6.7090	22.5462
final_refit_pred_24h.csv	1,992	pred	0.0346	221.4378	7.5388	25.8260
final_refit_pred_48h.csv	3,984	pred	0.0282	443.0379	8.7018	32.7261

The predictions folder also contains 8 pair_*_val_predictions.csv file(s) with row-level validation outputs for candidate analysis.

Demo-only evaluation files

These metrics were computed against any demo labels bundled with the repository. They are useful for quick checks, but they are not the official hidden-test challenge results.

file	overall_rmse	overall_mae	avg_county_rmse	n_rows	n_locations
demo_eval_24h.json	24.1382	6.5590	6.8424	1,992	83
demo_eval_48h.json	23.3941	6.2895	6.6395	3,984	83
demo_eval_final_refit_24h.json	26.7626	7.1971	7.6825	1,992	83
demo_eval_final_refit_48h.json	33.7444	8.3644	9.2266	3,984	83

Reusable model artifacts

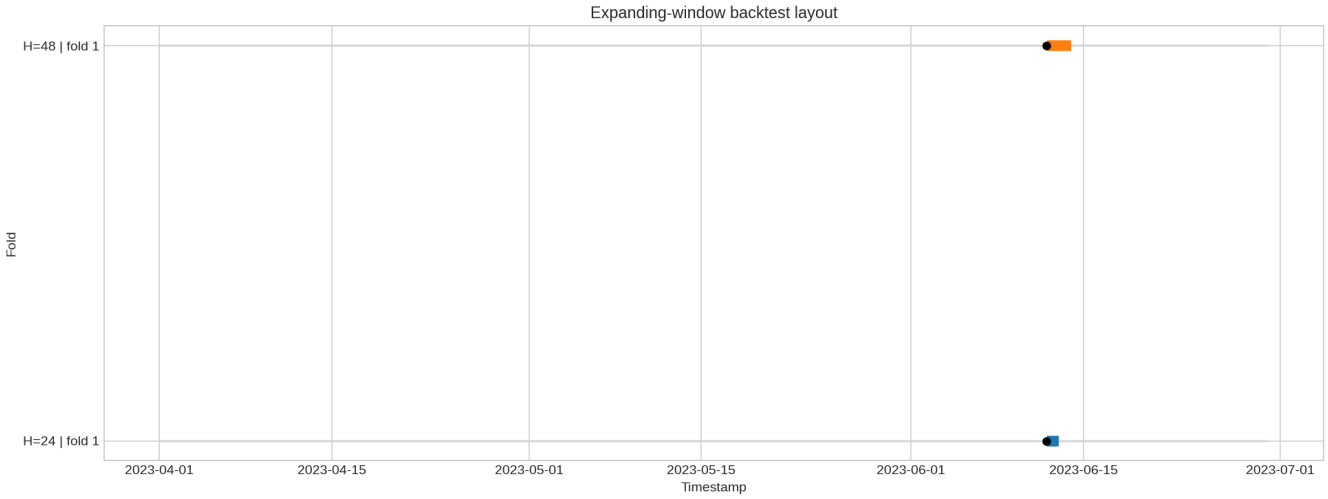
horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
24	1	h24__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	pair_h24__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib	pair_h24__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h24__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib

horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
48	1	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	pair_h48__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib

Visual appendix

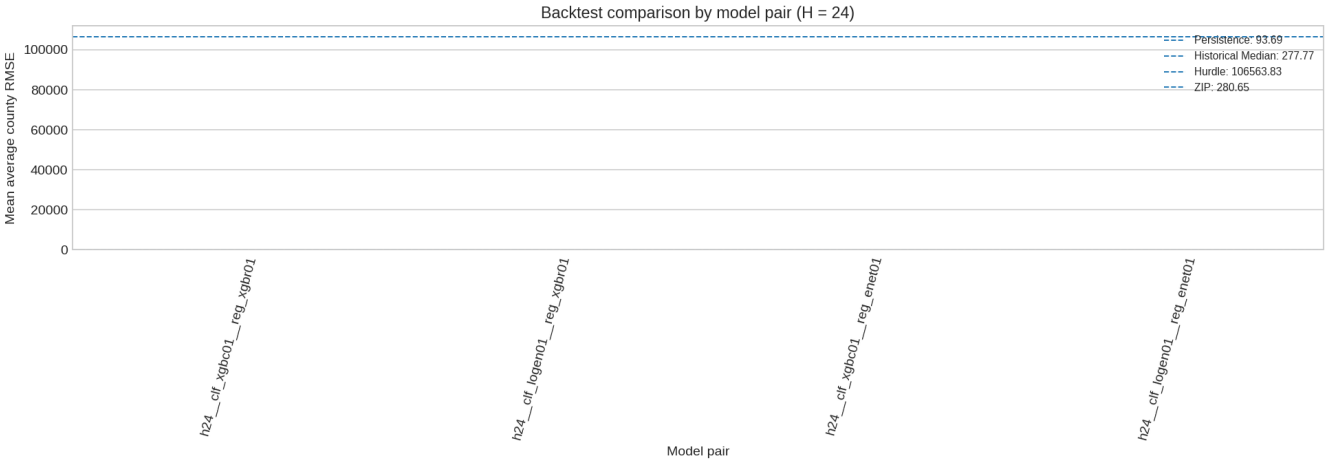
The following figures were read directly from the run's plots folder and inserted without GenAI interpretation.

Backtest Layout



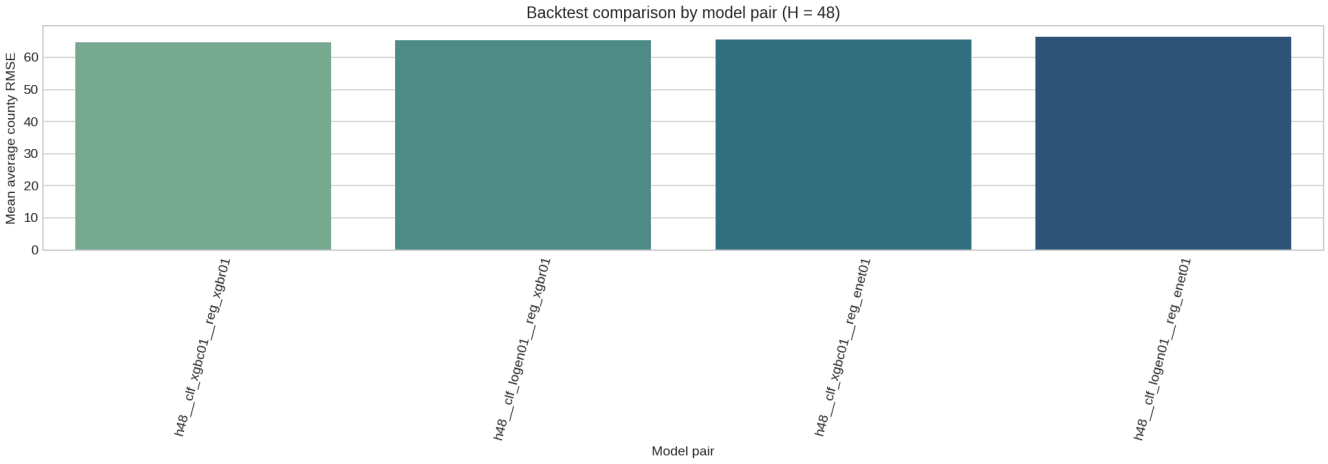
backtest_layout.png

Pair Comparison 24H



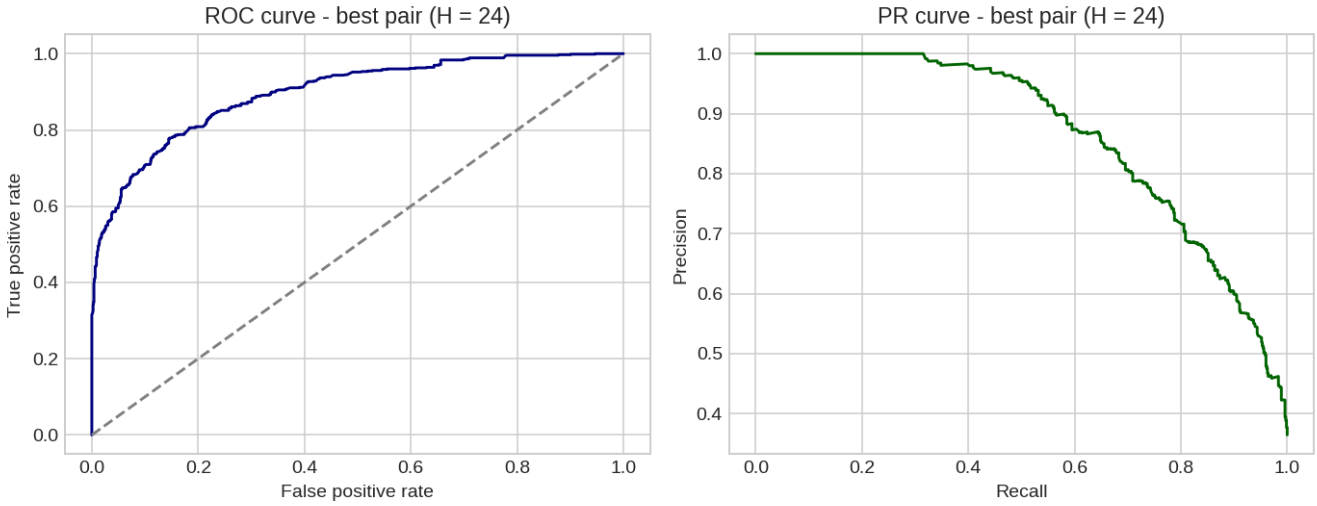
pair_comparison_h24.png

Pair Comparison 48H



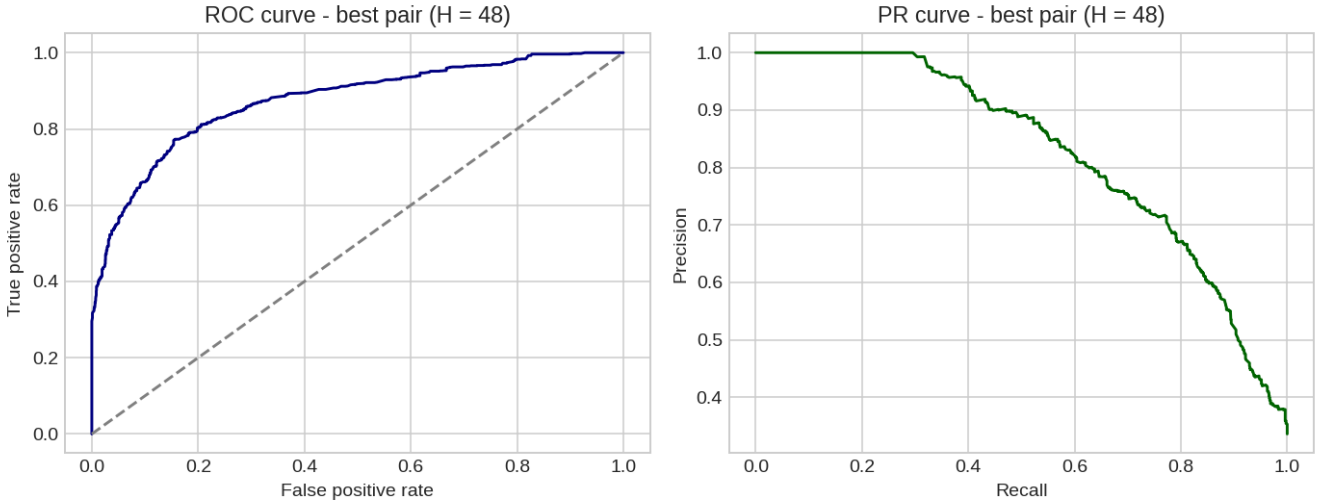
pair_comparison_h48.png

Best Pair Curves 24H



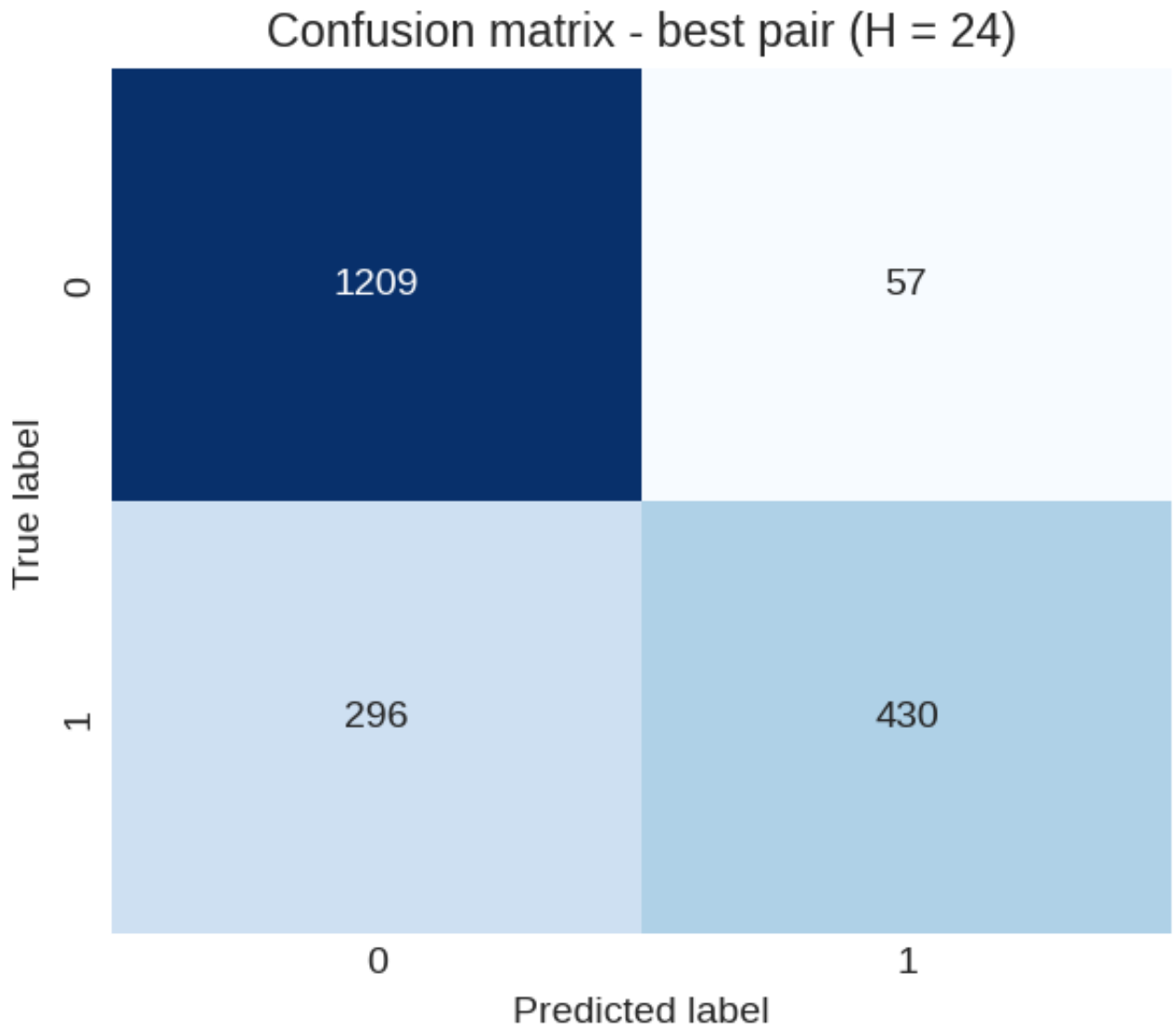
best_pair_curves_h24.png

Best Pair Curves 48H



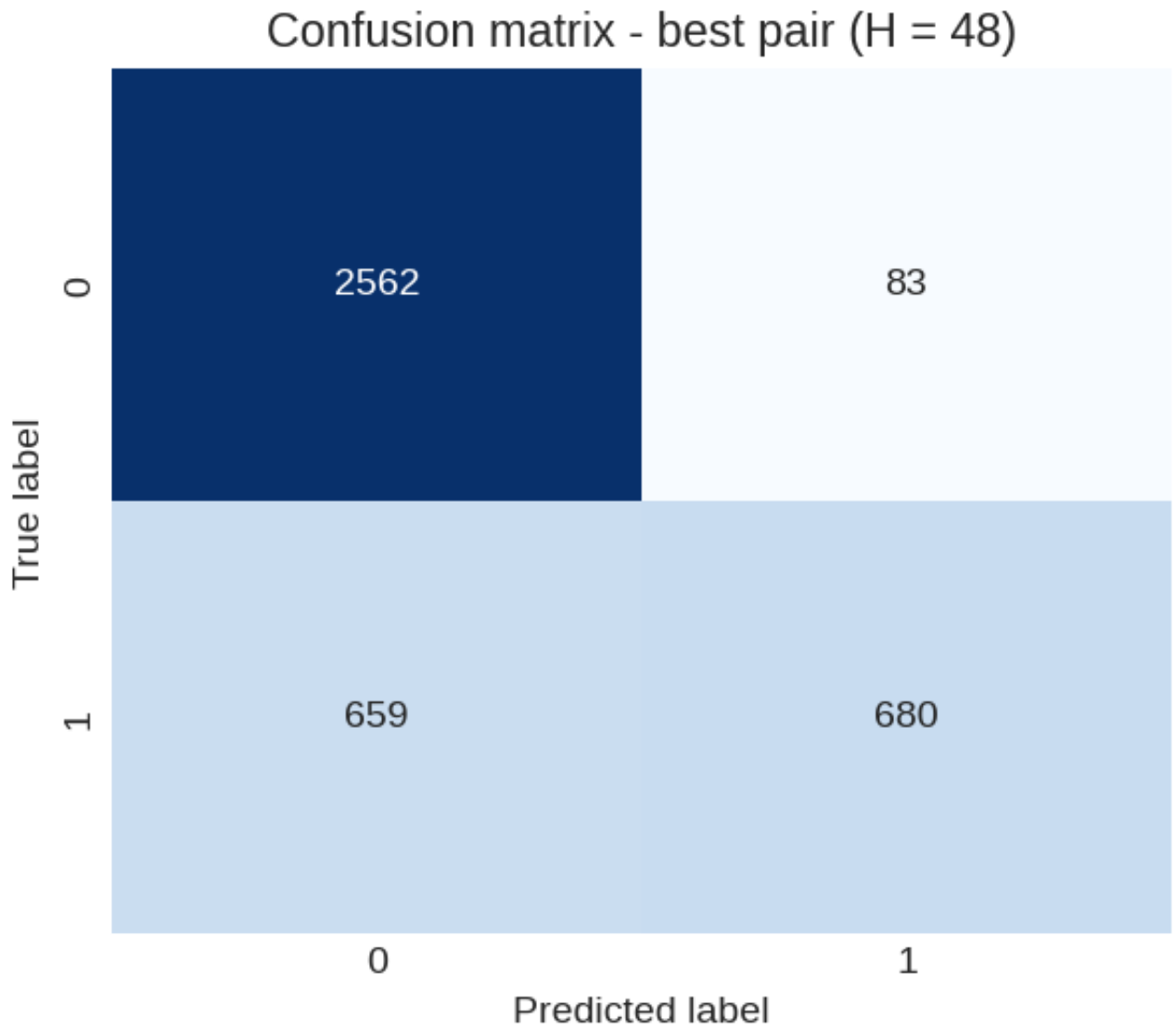
best_pair_curves_h48.png

Best Pair Confusion 24H



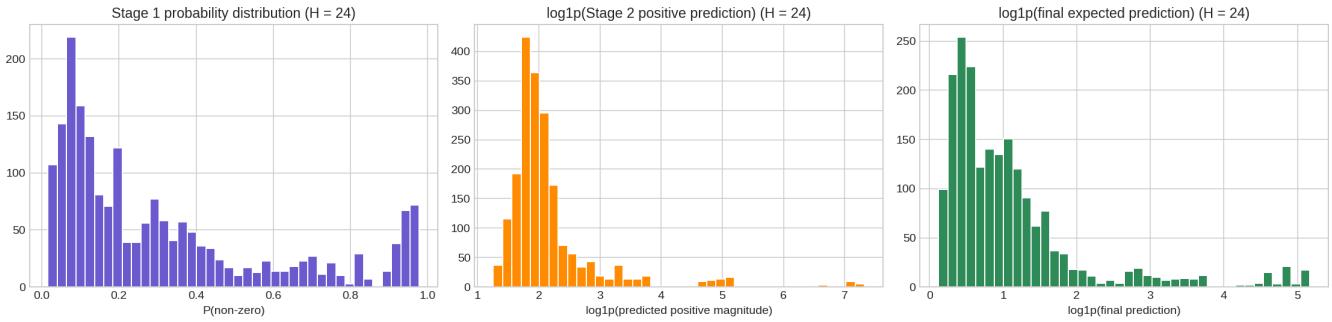
best_pair_confusion_h24.png

Best Pair Confusion 48H



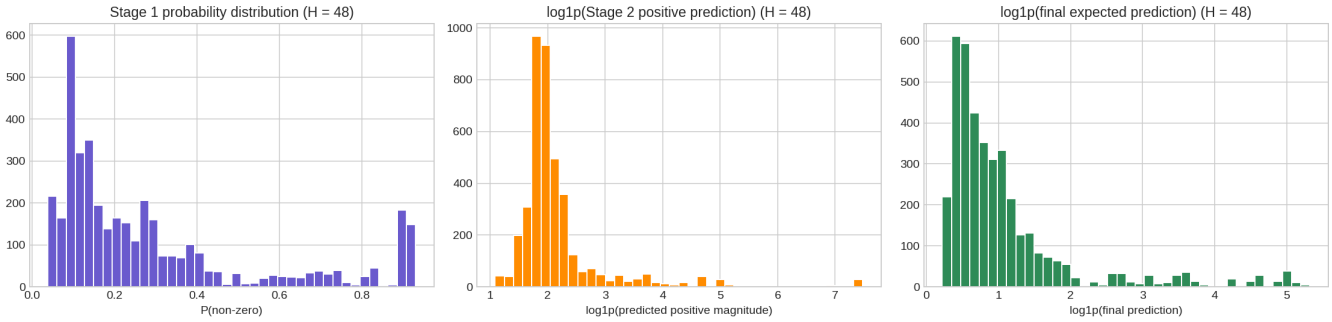
best_pair_confusion_h48.png

Prediction Distributions 24H



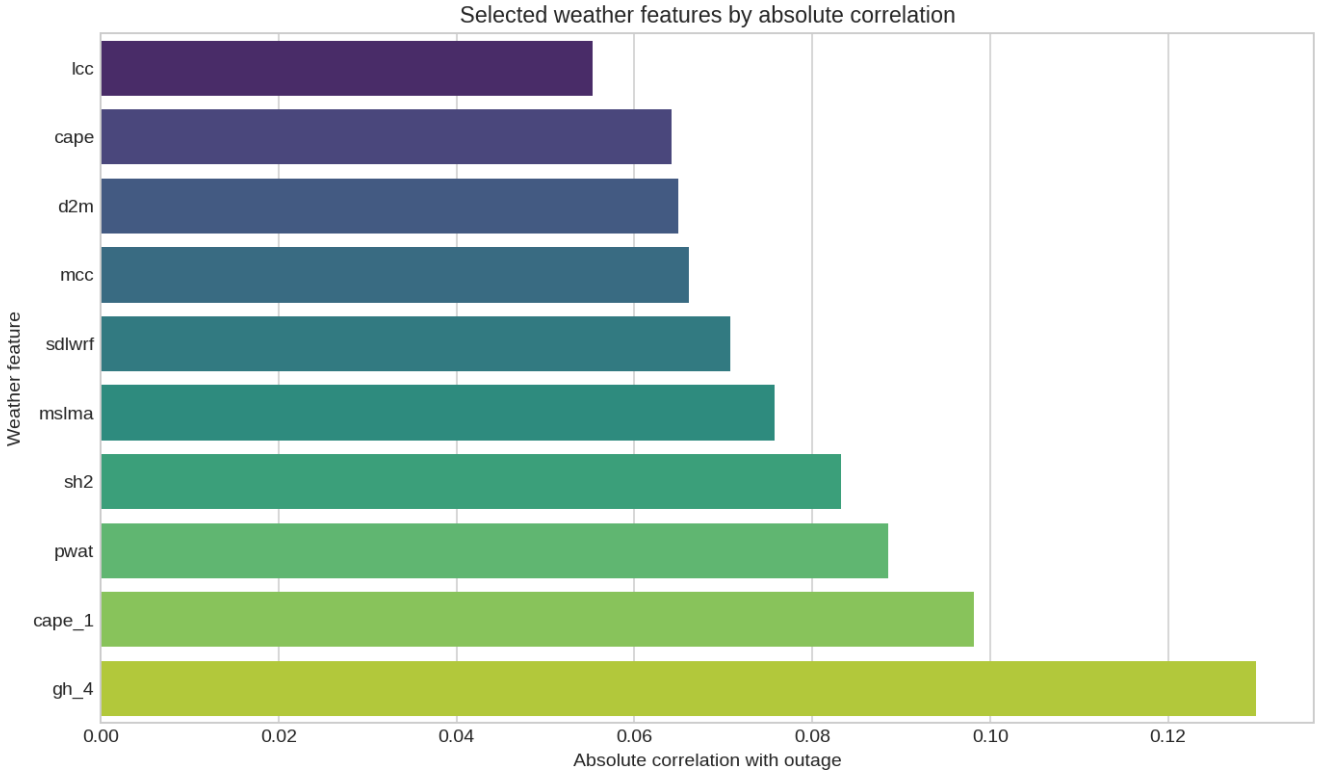
prediction_distributions_h24.png

Prediction Distributions 48H



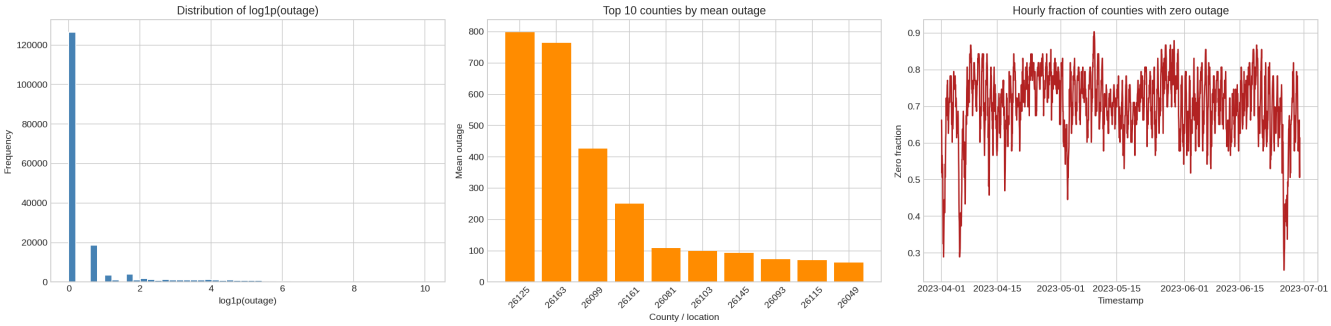
prediction_distributions_h48.png

Selected Weather Features



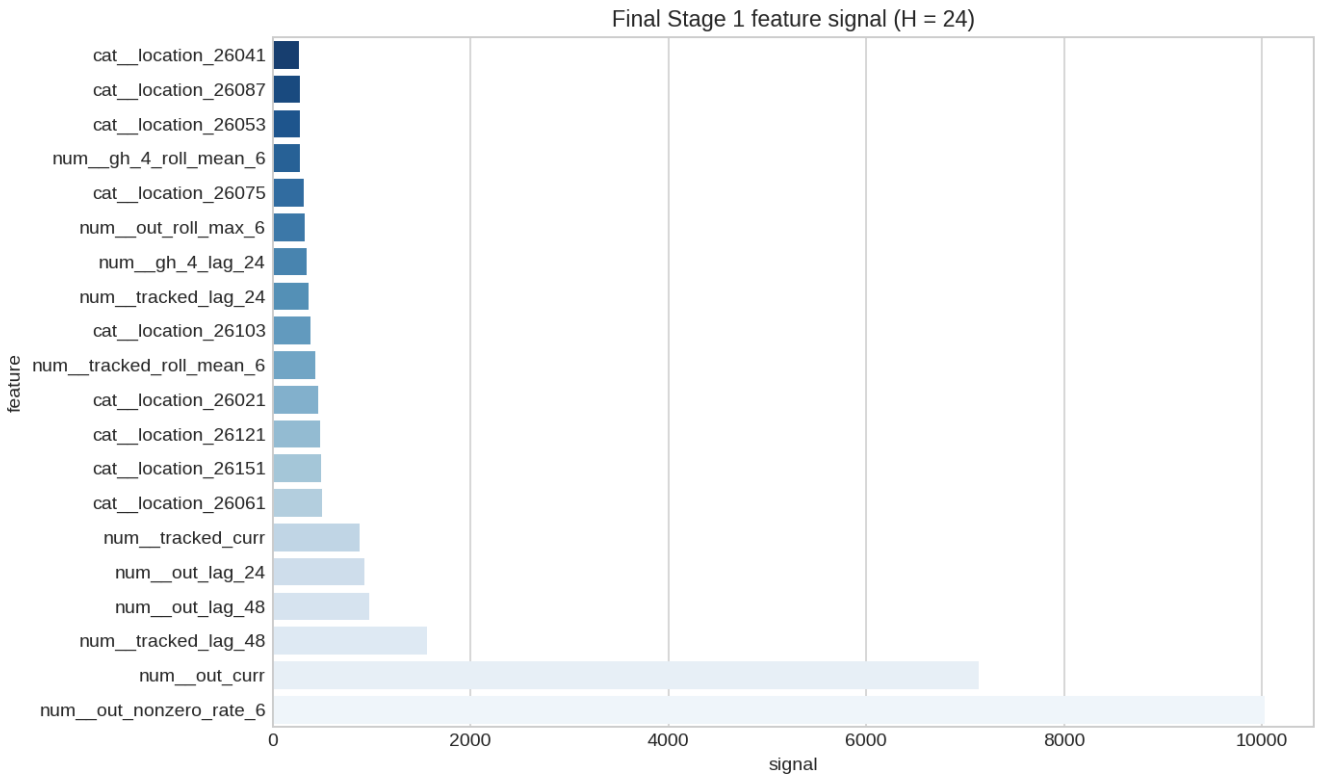
selected_weather_features.png

Eda Overview



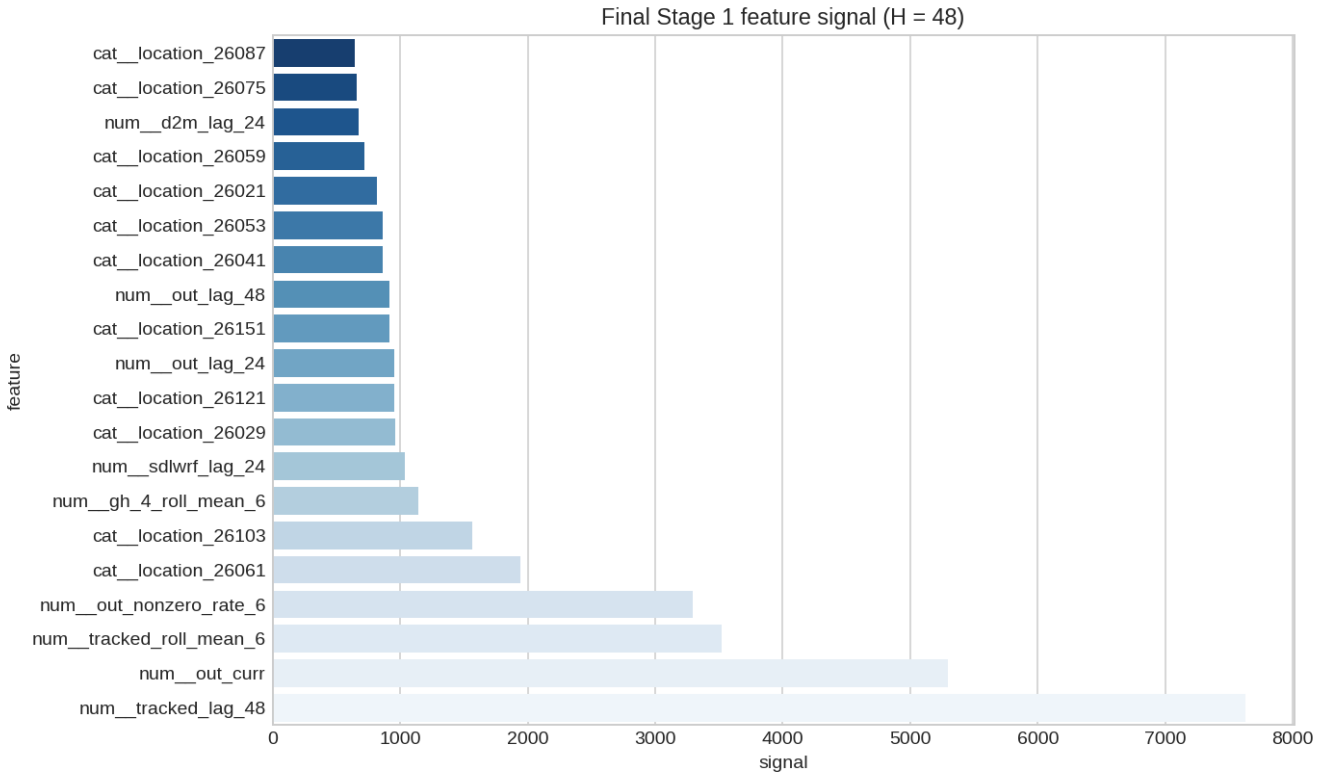
eda_overview.png

Final Stage1 Feature Signal 24H



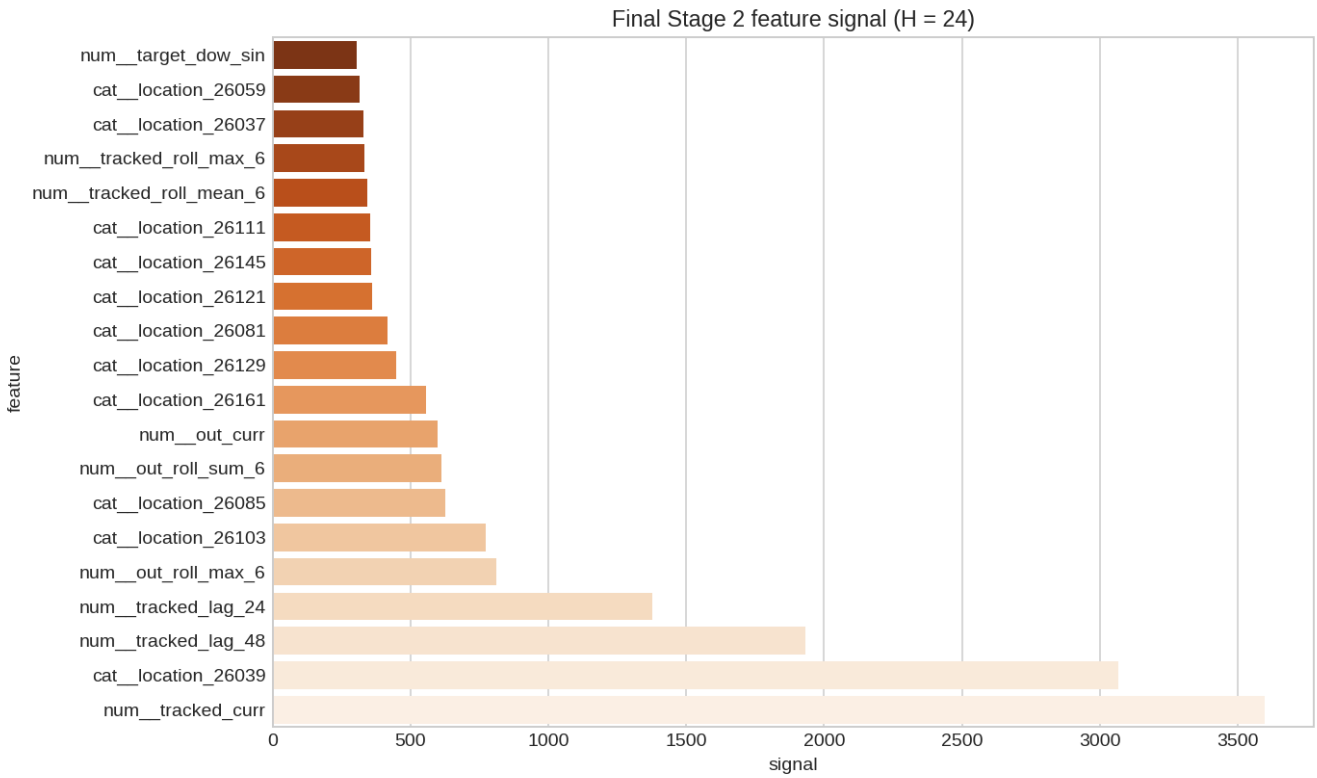
final_stage1_feature_signal_h24.png

Final Stage1 Feature Signal 48H



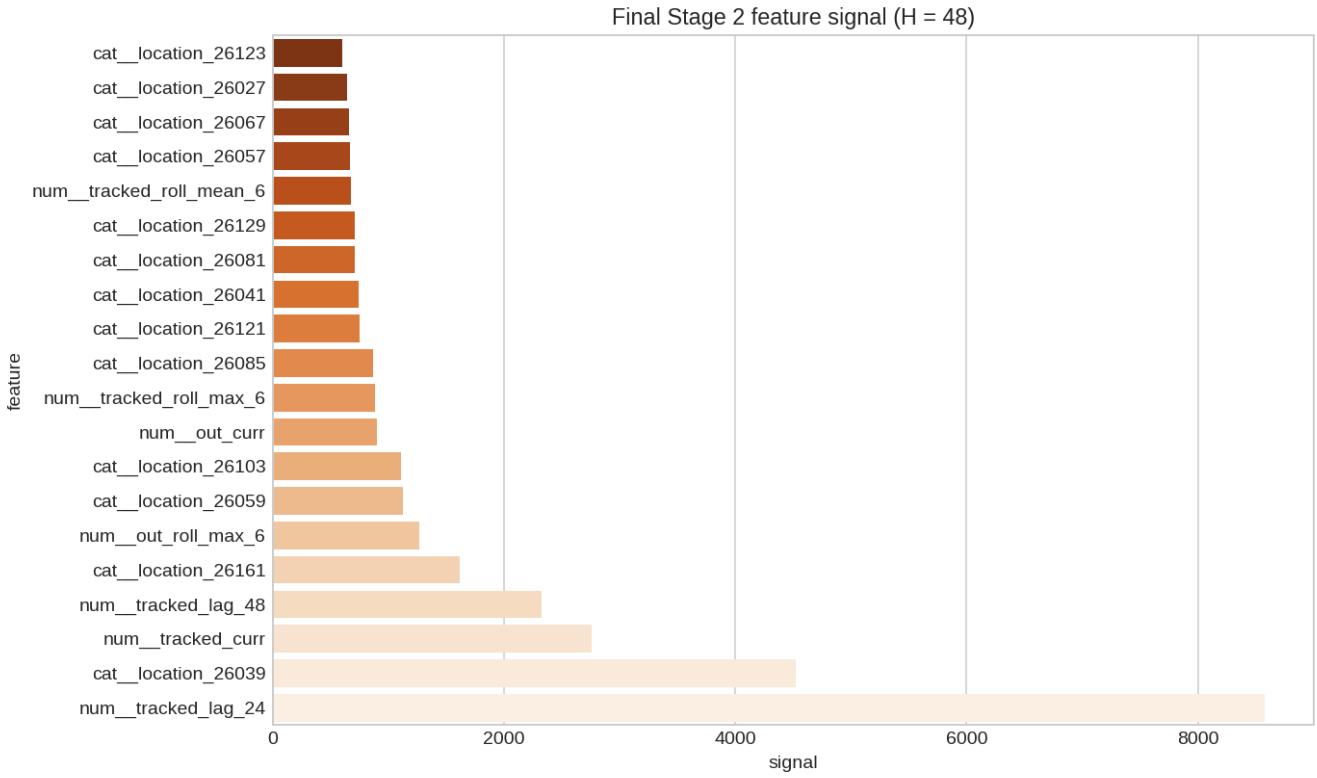
final_stage1_feature_signal_h48.png

Final Stage2 Feature Signal 24H



final_stage2_feature_signal_h24.png

Final Stage2 Feature Signal 48H

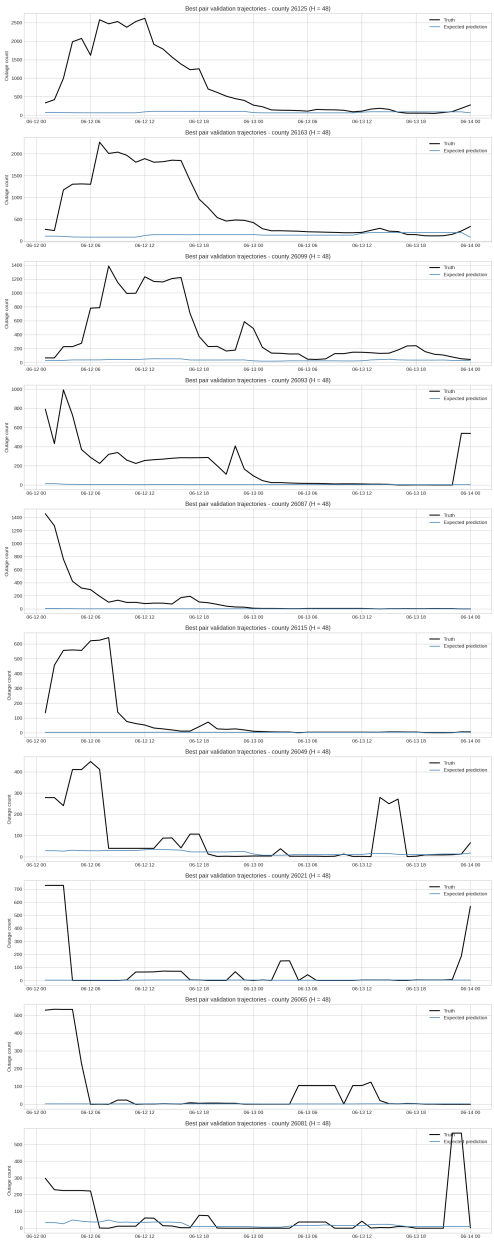


final_stage2_feature_signal_h48.png

Best Pair Validation Trajectories 24H

best_pair_validation_trajectories_h24.png

Best Pair Validation Trajectories 48H



best_pair_validation_trajectories_h48.png

Folder guide

- tables/: aggregated summaries, rankings, split definitions, and run-level CSV records.
- predictions/: exported submission-style predictions and detailed candidate validation predictions.
- models/: reusable .joblib artifacts plus metadata JSON files for submission and refit bundles.
- metrics/: fold-level metrics, training histories, and optional demo evaluation JSON files.
- plots/: visual diagnostics such as split layouts, model comparisons, confusion matrices, and feature-signal plots.
- checkpoints/: runtime resume artifacts used during notebook execution. This folder may be intentionally absent from GitHub uploads.